

# IDEALS ATTAINING EQUALITY IN THE DEMAILLY–PHAM INEQUALITY

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**ABSTRACT.** Let  $I \subset \mathbb{C}[[x_1, \dots, x_n]]$  have height at least  $l$ . We prove that  $\text{lct}(I) = E_l(I)$  if and only if the integral closure of  $I$  is the integral closure of  $(y_1^{d_1}, \dots, y_l^{d_l})$  in a regular system of parameters, as conjectured by Baily. The primary case follows from a simultaneous induction for ideals and divisorial valuations. The passage to arbitrary codimension uses transverse primary truncations and compatible finite-order coordinate changes. For ideals of convergent power series, Artin approximation gives holomorphic coordinates and a classification, modulo bounded terms, of plurisubharmonic analytic singularities attaining equality. We also prove finite generation of the graded ring of every finite-discrepancy real valuation satisfying  $\text{DP}(v) = A(v)$ , and apply this to the minimizers of  $(A(v) + v(\mathfrak{q}))^n \text{vol}(v)$ .

## 1. INTRODUCTION

Let  $\widehat{R} = \mathbb{C}[[x_1, \dots, x_n]]$ , with maximal ideal  $\widehat{\mathfrak{m}}$ . For an ideal  $I$  of height at least  $l$ , set

$$\sigma_j(I) = \sup_{a \geq 1} e((I + \widehat{\mathfrak{m}}^a)^{[j]}, \widehat{\mathfrak{m}}^{[n-j]}) \quad (1 \leq j \leq l),$$

$$\sigma_0(I) = 1, \quad E_l(I) = \sum_{j=0}^{l-1} \frac{\sigma_j(I)}{\sigma_{j+1}(I)}.$$

These extended mixed multiplicities are finite. Equivalently,  $\sigma_j(I)$  is the multiplicity of  $I$  on a sufficiently general  $j$ -dimensional linear section. The Demailly–Pham inequality [8], applied on a general  $l$ -dimensional section, gives

$$(1.1) \quad E_l(I) \leq \text{lct}(I).$$

The equality case concerns the integral closure of  $I$ , since both sides are invariant under integral closure. Bivià-Ausina [3] characterized primary diagonal ideals using equality and a condition on the containing monomial ideal. Baily [2, Theorem 5.1] classified homogeneous equality ideals and

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formulated the corresponding local classification in [2, Conjecture 6.2]. Our main result is this local classification in every codimension. For convergent ideals it also gives the analytic-coordinate classification proposed in [4, Remark 3.7] in the primary case.

**Theorem 1.1** (Equality ideals in arbitrary codimension). *Let  $0 \neq I \subsetneq \widehat{R} = \mathbb{C}[[x_1, \dots, x_n]]$  and  $1 \leq l \leq \text{ht } I$ . Then  $\text{lct}(I) = E_l(I)$  if and only if there exist a regular system of parameters  $y_1, \dots, y_n$  of  $\widehat{R}$  and positive integers  $d_1 \leq \dots \leq d_l$  such that*

$$(1.2) \quad \bar{I} = \overline{(y_1^{d_1}, \dots, y_l^{d_l})}.$$

In this case  $\text{ht } I = l$  and

$$(1.3) \quad d_j = \frac{\sigma_j(I)}{\sigma_{j-1}(I)} \quad (1 \leq j \leq l), \quad \text{lct}(I) = \sum_{j=1}^l \frac{1}{d_j}.$$

The parameters belong to the original complete local ring, not to a higher birational model. They are formal; no assertion of convergence is needed for this statement. In particular, equality forces the reduced support of  $I$  to be smooth. Section 5 proves a separate reduction theorem showing that the primary classification in dimension  $n$  implies the classification for every  $l \leq n$  in that same dimension.

**1.1. The primary case and divisorial rigidity.** For the valuative arguments put  $R = \mathbb{C}[x_1, \dots, x_n]_{(x_1, \dots, x_n)}$ , with maximal ideal  $\mathfrak{m}$ , and let  $o$  be its closed point. Unless specified otherwise,  $n \geq 2$  in these arguments. For a primary ideal  $K$ , write

$$e_j(K) = e(K^{[j]}, \mathfrak{m}^{[n-j]}), \quad e_0(K) = 1, \quad \text{DP}(K) = E_n(K).$$

For a real valuation  $v$  centered at  $o$ , let  $A(v)$  denote its log discrepancy and put

$$\mathfrak{a}_t(v) = \{f \in R : v(f) \geq t\}, \quad \text{vol}(v) = \lim_{m \rightarrow \infty} \frac{n! \ell(R/\mathfrak{a}_m(v))}{m^n}.$$

When  $A(v) < \infty$ , the asymptotic mixed multiplicities

$$(1.4) \quad b_0(v) = 1, \quad b_j(v) = \lim_{m \rightarrow \infty} \frac{e_j(\mathfrak{a}_m(v))}{m^j} \quad (1 \leq j \leq n)$$

exist and are positive, with  $b_n(v) = \text{vol}(v)$ . Set

$$\text{DP}(v) = \sum_{j=0}^{n-1} \frac{b_j(v)}{b_{j+1}(v)}.$$

The finite-level inequality implies

$$(1.5) \quad \text{DP}(v) \leq \text{lct}(\mathfrak{a}_\bullet(v)) \leq A(v).$$

Section 2 gives the conventions and proofs needed for these asymptotic statements.

A valuation is monomial in regular parameters  $z_1, \dots, z_n$  if the value of a nonzero series  $\sum_{\beta} c_{\beta} z^{\beta}$  is  $\min_{c_{\beta} \neq 0} \sum_i \beta_i v(z_i)$ . The parameters in the following theorem belong to the original completed local ring, not to a higher birational model.

**Theorem 1.2** (Primary classification and divisorial rigidity).

- (i) *An  $\mathfrak{m}$ -primary ideal  $I \subset R$  satisfies  $\text{DP}(I) = \text{lct}(I)$  if and only if there are regular parameters  $z_1, \dots, z_n$  of  $\widehat{R}$  and positive integers  $d_1, \dots, d_n$  such that*

$$(1.6) \quad \overline{I\widehat{R}} = \overline{(z_1^{d_1}, \dots, z_n^{d_n})}.$$

*The same assertion holds for  $\widehat{\mathfrak{m}}$ -primary ideals of  $\widehat{R}$ , where  $\widehat{\mathfrak{m}} = \mathfrak{m}\widehat{R}$ .*

- (ii) *If  $v$  is divisorial, centered at  $o$ , and  $\text{DP}(v) = A(v)$ , then its extension to  $\widehat{R}$  is monomial in regular parameters. In particular,  $\text{gr}_v R$  is a weighted polynomial ring.*

The two assertions are proved together, without using Theorem 1.1. Part (i) is the primary input to that theorem. Part (ii) gives the closed-point divisorial case of the valuative formulation in [2, Conjecture 6.3].

**1.2. Finite generation for real valuations.** For an arbitrary real valuation, the equality in (1.5) has a finite-generation consequence. Write

$$\text{gr}_v R = \bigoplus_{\lambda \in \mathbb{R}_{\geq 0}} \mathfrak{a}_{\lambda}(v) / \mathfrak{a}_{>\lambda}(v), \quad \mathfrak{a}_{>\lambda}(v) = \{f : v(f) > \lambda\}.$$

**Theorem 1.3** (Finite generation). *Let  $v$  be a real valuation centered at  $o$ , with  $A(v) < \infty$ . If  $\text{DP}(v) = A(v)$ , then  $v$  is quasi-monomial and special. Consequently  $\text{gr}_v R$  is a finitely generated  $\mathbb{C}$ -algebra and  $\text{Spec}(\text{gr}_v R)$  is klt.*

Special valuations are used in the sense of Xu–Zhuang [17]. We apply their finite-generation theorem through Chen’s boundary-perturbation criterion [6, Lemma 3.10]. The criterion and the relevant terminology are recalled in Section 6. Theorem 1.3 does not identify a general real equality valuation as monomial in parameters of  $\widehat{R}$ .

**1.3. An ideal-weighted volume.** Fix a nonzero ideal  $\mathfrak{q} \subseteq R$  and consider

$$(1.7) \quad F_{\mathfrak{q}}(v) = (A(v) + v(\mathfrak{q}))^n \text{vol}(v), \quad A(v) < \infty.$$

The functional is invariant under positive rescaling of  $v$ . We allow  $\mathfrak{q} = R$ , with  $v(R) = \text{ord}_o(R) = 0$ .

**Theorem 1.4** (Minimum and equality profile). *Let  $d = \text{ord}_o(\mathfrak{q})$  and  $D = d + 1$ . Then*

$$\min_{v : A(v) < \infty} F_{\mathfrak{q}}(v) = n^n D.$$

Every minimizer satisfies  $\text{DP}(v) = A(v)$ , and hence has a finitely generated associated graded ring with klt spectrum. Under the normalization  $v(\mathfrak{m}) = 1$ , every minimizer satisfies

$$(1.8) \quad \begin{aligned} v(\mathfrak{q}) &= d, & A(v) &= 1 + (n-1)D, \\ b_j(v) &= D^{-j} \quad (1 \leq j < n), & b_n(v) &= D^{-(n-1)}. \end{aligned}$$

The minimum is attained by a monomial valuation of weights  $(1, D, \dots, D)$  in suitable linear coordinates. Theorem 1.2 further identifies every divisorial minimizer with a valuation of these weights in formal regular parameters, subject to  $v(\mathfrak{q}) = d$ ; see Corollary 7.2. For  $\mathfrak{q} \neq R$ , (1.7) is not the ordinary normalized volume of an effective klt pair, and no K-semistability assertion is made for its minimizers.

**1.4. Plurisubharmonic analytic singularities.** For a plurisubharmonic germ  $\varphi$ , let  $c_o(\varphi) = \sup\{c > 0 : e^{-2c\varphi} \in L^1_{\text{loc}}(o)\}$ . We use  $dd^c = (i/\pi)\partial\bar{\partial}$  and write  $e_j(\varphi) = \nu((dd^c\varphi)^j, o)$ , with  $e_0(\varphi) = 1$ . For analytic singularities whose zero set has codimension at least  $l$ , these numbers are defined for  $j \leq l$ ; precise conventions are given in Section 8.2. The inequality  $c_o(\varphi) \geq \sum_{j=1}^l e_{j-1}(\varphi)/e_j(\varphi)$  is due to Demailly–Pham for isolated singularities [8] and to Rashkovskii in this codimension range [14, Theorem 1.4].

**Corollary 1.5** (Analytic singularities attaining equality). *Let  $\varphi$  be a plurisubharmonic germ at  $o \in \mathbb{C}^n$  of the form*

$$\varphi = \frac{\lambda}{2} \log \left( \sum_{\alpha=1}^s |f_\alpha|^2 \right) + O(1), \quad \lambda > 0,$$

where  $0 \neq I = (f_1, \dots, f_s) \subsetneq \mathcal{O}_{\mathbb{C}^n, o}$  and  $1 \leq l \leq \text{codim}_o V(I)$ . Then

$$(1.9) \quad c_o(\varphi) = \sum_{j=1}^l \frac{e_{j-1}(\varphi)}{e_j(\varphi)}$$

if and only if there are holomorphic coordinates  $z_1, \dots, z_n$  at  $o$  and positive integers  $d_1 \leq \dots \leq d_l$  such that

$$(1.10) \quad \varphi = \lambda \max_{1 \leq j \leq l} \{d_j \log |z_j|\} + O(1).$$

In this case  $V(I)_{\text{red}} = \{z_1 = \dots = z_l = 0\}$  and

$$(1.11) \quad e_j(\varphi) = \lambda^j d_1 \cdots d_j \quad (1 \leq j \leq l), \quad c_o(\varphi) = \frac{1}{\lambda} \sum_{j=1}^l \frac{1}{d_j}.$$

This is a classification of singularity types modulo bounded terms, not of individual plurisubharmonic functions. The case  $l = n$  classifies weights with isolated analytic singularities. The proof in Section 8 first replaces formal coordinates by holomorphic ones, using finite systems of integral-dependence equations and Artin approximation [1]. It then identifies integral closure with the required bounded comparison of logarithms.

**1.5. Method of proof and organization.** Two estimates connect the ideal and valuative statements. First, the DP function is nonincreasing on coordinatewise increasing segments of the positive log-convex cone; equality for two comparable vectors forces equality of all coordinates. Second, for any nonzero  $f \in \mathfrak{m}$ ,

$$(1.12) \quad \lim_{m \rightarrow \infty} \frac{e(\mathfrak{a}_m(v); R/(f))}{m^{n-1}} = v(f) \operatorname{vol}(v).$$

The latter identity follows from the exact colon formula and Cutkosky's multiplicity theorem [7]. These inputs are established in Sections 2 and 3.

Section 4 proves Theorem 1.2 by simultaneous induction. An equality ideal determines a unique normalized divisorial valuation computing its threshold. Blum's results [5] give its finite generation and a plt extraction. A general hyperplane preserves the equality, and [17, Lemma 2.17] controls its Rees degeneration. Normality of the quotient Rees algebra and reducedness of its central fiber allow us to identify the entire quotient filtration and lift monomial parameters from dimension  $n - 1$  to dimension  $n$ .

Section 5 passes from primary ideals to arbitrary codimension. If  $B$  is generated by  $n - l$  sufficiently general linear forms, comparison of mixed multiplicities gives

$$E_n(I + B^N) \geq E_l(I) + \frac{n - l}{N} \quad (N \gg 0).$$

The upper bound for  $\operatorname{lct}(I + B^N)$  obtained from the sum of ideals is the same when  $I$  is an equality ideal. Thus every sufficiently high transverse truncation is a primary equality ideal. Its diagonal exponents are fixed by the mixed multiplicities, except for the  $n - l$  exponents equal to  $N$ . Integral-closure continuity then produces coordinate equivalences to every finite order. We pass to compatible jets by the stabilization of algebraic subgroup cosets; this step does not assume that the coordinate changes initially supplied by the primary theorem are compatible.

Section 6 treats real valuations. The two estimates above give a uniform discrepancy gap under domination. This implies uniqueness of the valuation computing its own threshold, quasi-monomiality by [16], and stability under small effective boundary perturbations. The special-valuation criterion then proves Theorem 1.3. Section 7 proves Theorem 1.4 and classifies its divisorial minimizers. Section 8 proves the analytic classification and Corollary 1.5.

## 2. MIXED MULTIPLICITIES, THRESHOLDS, AND VALUATIONS

For the algebraic valuation arguments,  $R = \mathbb{C}[x_1, \dots, x_n]_{(x_1, \dots, x_n)}$  and  $\mathfrak{m} = (x_1, \dots, x_n)$ . The arbitrary-codimension arguments take place in  $\widehat{R} = \mathbb{C}[[x_1, \dots, x_n]]$ . All valuations used in the valuative sections are finite on nonzero elements and centered at  $\mathfrak{m}$ . For a nonzero ideal  $I$ ,  $v(I)$  is the minimum of the values of any finite generating set. We write  $e(I; S)$  for the

Hilbert–Samuel multiplicity of an ideal of finite colength in a local ring  $S$  and  $\bar{I}$  for integral closure.

**2.1. Ideals of positive-dimensional support.** We recall the facts needed for the transverse truncation argument. They apply to both  $R$  and  $\hat{R}$ ; in this subsection write  $(S, \mathfrak{n})$  for either ring. For ideals  $I_1, \dots, I_n$  set

$$\sigma(I_1, \dots, I_n) = \sup_{a \geq 1} e(I_1 + \mathfrak{n}^a, \dots, I_n + \mathfrak{n}^a).$$

It is finite if one can choose  $g_i \in I_i$  whose common ideal is  $\mathfrak{n}$ -primary. In that case it equals the multiplicity of  $(g_1, \dots, g_n)$  for general choices, and is no larger than that multiplicity for any such parameter tuple. It decreases when any of its ideal arguments is enlarged. See [10, Chapter 17] and [2, Proposition 2.37]. In particular,

$$\sigma_j(I) = \sigma(I^{[j]}, \mathfrak{n}^{[n-j]}), \quad \sigma_j(I)^2 \leq \sigma_{j-1}(I)\sigma_{j+1}(I)$$

whenever these quantities are finite. Finiteness holds exactly for  $j \leq \text{ht } I$ .

For  $r = n - l$ , choose general linear forms  $z_1, \dots, z_r$  and put  $S_L = S/(z_1, \dots, z_r)$ . The image  $IS_L$  has finite colength and multiplicity  $\sigma_l(I)$ . Choose  $l$  general elements of  $I$  whose images generate a reduction of  $IS_L$ . Since  $S_L$  is Cohen–Macaulay, their parameter-ideal multiplicity is their colength. Thus there are  $f_1, \dots, f_l \in I$  with

$$(2.1) \quad \ell(S/(f_1, \dots, f_l, z_1, \dots, z_r)) = \sigma_l(I).$$

This tuple is a regular sequence. Replacing any of the  $z_i$  by its  $N$ -th power multiplies its colength by  $N$ .

We use the threshold identities and inequalities

$$(2.2) \quad \text{lct}(I) = \text{lct}(\bar{I}), \quad \text{lct}(I + J) \leq \text{lct}(I) + \text{lct}(J), \quad \text{lct}(B^N) = \frac{r}{N}$$

when  $B$  is generated by  $r$  independent linear forms. Restriction to a smooth section cannot increase the threshold. These facts hold for formal ideals as well; see [2, Proposition 2.10] and the valuative treatment in [11]. Restricting to a general  $l$ -dimensional section and applying [8] proves  $E_l(I) \leq \text{lct}(I)$ .

**2.2. Valuations and linear bounds.** We use  $v(0) = +\infty$  and  $v(c) = 0$  for  $c \in \mathbb{C}^*$ . For  $t \leq 0$  set  $\mathfrak{a}_t(v) = R$ . The relation  $w \geq v$  means  $w(f) \geq v(f)$  for every  $f \in R$ . Log discrepancy and quasi-monomial valuations are used with the conventions of [11, Sections 3 and 5]. In particular, on a log smooth model, log discrepancy is the weighted sum of the discrepancies of the divisors defining a quasi-monomial valuation.

**Lemma 2.1.** *If  $s = v(\mathfrak{m}) > 0$  and  $A = A(v) < \infty$ , then*

$$s \text{ord}_o(f) \leq v(f) \leq A \text{ord}_o(f) \quad (0 \neq f \in \mathfrak{m}).$$

*Consequently, for  $t > 0$ ,*

$$(2.3) \quad \mathfrak{m}^{\lceil t/s \rceil} \subseteq \mathfrak{a}_t(v) \subseteq \mathfrak{m}^{\lceil t/A \rceil}.$$

*Proof.* The first inequality follows from  $f \in \mathfrak{m}^{\text{ord}_o(f)}$ . For the other inequality, Skoda's bound gives  $\text{lct}(f) \geq 1/\text{ord}_o(f)$ , while the valuative formula for the threshold gives  $\text{lct}(f) \leq A(v)/v(f)$ . Combining the two proves the assertion. The Skoda bound and its normalization are recalled in [8, Section 1]; the valuative formula is treated in [11]. The ideal inclusions follow by applying the inequalities to each element.  $\square$

### 2.3. Asymptotic mixed multiplicities.

**Theorem 2.2** (Cutkosky). *Let  $(S, \mathfrak{n})$  be an analytically unramified equicharacteristic local ring of dimension  $r > 0$  with perfect residue field. For a graded family  $J_\bullet$  of  $\mathfrak{n}$ -primary ideals,*

$$\lim_{m \rightarrow \infty} \frac{r! \ell(S/J_m)}{m^r} = \lim_{m \rightarrow \infty} \frac{e(J_m; S)}{m^r}.$$

*Both limits exist.*

This is [7, Theorems 5.8 and 6.5]. Reduced local rings essentially of finite type over  $\mathbb{C}$ , and reduced complex analytic local rings, satisfy the hypotheses.

**Lemma 2.3.** *For  $v$  as in Lemma 2.1, all  $b_j(v)$  exist and*

$$A(v)^{-j} \leq b_j(v) \leq s^{-j}, \quad b_j(v)^2 \leq b_{j-1}(v)b_{j+1}(v) \quad (1 \leq j < n).$$

*Moreover  $b_n(v) = \text{vol}(v)$ , and  $w \geq v$  implies  $b_j(w) \leq b_j(v)$  whenever both discrepancies are finite.*

*Proof.* For each  $j$ , choose  $n - j$  sufficiently general linear forms such that, on the resulting regular local quotient  $S_j$  of dimension  $j$ ,

$$e_j(\mathfrak{a}_m(v)) = e(\mathfrak{a}_m(v)S_j; S_j) \quad \text{for every integer } m \geq 1.$$

Such a choice is possible: the general-section formula for mixed multiplicities imposes a nonempty Zariski open condition for each fixed ideal, and there are only countably many ideals. Over the uncountable field  $\mathbb{C}$ , these conditions can be satisfied simultaneously. See [10, Chapter 17] for the general-section formula. The images of the valuation ideals form a graded primary family on  $S_j$ . Theorem 2.2 proves existence of the required limit. For  $j = n$  it identifies that limit with volume.

Inclusions (2.3), monotonicity of mixed multiplicities, and  $e_j(\mathfrak{m}^k) = k^j$  give the bounds. The finite-level inequalities  $e_j(I)^2 \leq e_{j-1}(I)e_{j+1}(I)$  pass to the limit; see [8, Lemma 2.1]. Finally  $w \geq v$  gives  $\mathfrak{a}_m(v) \subseteq \mathfrak{a}_m(w)$ . Mixed multiplicity decreases under enlargement of an ideal, which proves the last assertion.  $\square$

**2.4. Thresholds of graded sequences.** For a graded sequence  $I_\bullet$ , write  $u(I_\bullet) = \inf_{m \geq 1} u(I_m)/m$ . The asymptotic threshold has the descriptions

$$(2.4) \quad \text{lct}(I_\bullet) = \sup_{m \geq 1} m \text{lct}(I_m) = \inf_u \frac{A(u)}{u(I_\bullet)}.$$

Only valuations with positive denominator enter the infimum. For primary sequences with linear bounds they are centered at  $o$ . These formulas, and attainment by a valuation, are discussed in [11, Sections 6–7].

**Lemma 2.4.** *One has  $v(\mathfrak{a}_\bullet(v)) = 1$ . If  $w(\mathfrak{a}_\bullet(v)) \geq 1$ , then  $w \geq v$ . Furthermore,*

$$(2.5) \quad \text{DP}(v) \leq \text{lct}(\mathfrak{a}_\bullet(v)) \leq A(v), \quad A(v)^n \text{vol}(v) \geq n^n.$$

*Proof.* The definition gives  $v(\mathfrak{a}_m(v)) \geq m$ . Fix  $f \in \mathfrak{m} \setminus \{0\}$  and let  $m_k = \lfloor kv(f) \rfloor$ . Since  $f^{m_k} \in \mathfrak{a}_{m_k}(v)$ , the opposite bound follows on letting  $k \rightarrow \infty$ . If  $w(\mathfrak{a}_\bullet(v)) \geq 1$ , the same containment gives  $kw(f) \geq m_k$ , hence  $w(f) \geq v(f)$ .

The Demailly–Pham inequality for  $\mathfrak{a}_m(v)$  reads

$$m \text{lct}(\mathfrak{a}_m(v)) \geq \sum_{j=0}^{n-1} \frac{e_j(\mathfrak{a}_m(v))/m^j}{e_{j+1}(\mathfrak{a}_m(v))/m^{j+1}}.$$

Its limit, (2.4), and the preceding normalization give (2.5). The product of the  $n$  summands in  $\text{DP}(v)$  is  $1/b_n(v)$ . The arithmetic–geometric mean inequality therefore gives  $A(v) \geq \text{DP}(v) \geq n \text{vol}(v)^{-1/n}$ .  $\square$

**2.5. Numerical monotonicity.** Consider the positive log-convex region

$$\mathcal{D} = \{(t_1, \dots, t_n) > 0 : t_j^2 \leq t_{j-1}t_{j+1} \ (1 \leq j < n), \ t_0 = 1\}$$

and  $F(t) = \sum_{j=0}^{n-1} t_j/t_{j+1}$ . The following estimate makes the monotonicity in [3, Proposition 10] quantitative in its last coordinate.

**Lemma 2.5.** *The region  $\mathcal{D}$  is convex and  $F$  is nonincreasing along coordinatewise increasing line segments in  $\mathcal{D}$ . If  $q \leq p$  in  $\mathcal{D}$ , then  $F(q) = F(p)$  if and only if  $q = p$ . If  $p, q \in \mathcal{D}$ ,  $q_j \leq p_j$ , and  $F(p), F(q) \leq B$ , then*

$$(2.6) \quad F(q) - F(p) \geq \frac{p_n - q_n}{B^{n-1}p_n^2}.$$

*Proof.* For two log-convex sequences, Cauchy–Schwarz shows that their convex combination is log-convex: with  $z_j = (1-t)p_j + tq_j$ ,

$$z_j \leq (1-t)\sqrt{p_{j-1}p_{j+1}} + t\sqrt{q_{j-1}q_{j+1}} \leq \sqrt{z_{j-1}z_{j+1}}.$$

For  $1 \leq j < n$ , direct differentiation gives

$$\partial_j F = -\frac{t_{j-1}}{t_j^2} + \frac{1}{t_{j+1}} \leq 0, \quad \partial_n F = -\frac{t_{n-1}}{t_n^2} < 0.$$

Every summand of  $F(p)$  and  $F(q)$  is at most  $B$ , so induction gives  $p_j, q_j \geq B^{-j}$ . Along the segment  $z(t)$ ,  $z_{n-1} \geq B^{-(n-1)}$  and  $z_n \leq p_n$ . All contributions to  $\frac{d}{dt}F(z(t)) = \sum_j \partial_j F(z(t))(q_j - p_j)$  are nonnegative. Retaining the last one and integrating on  $[0, 1]$  proves (2.6).

To prove the equality assertion, suppose  $q \leq p$  and  $F(q) = F(p)$ . Along  $z(t) = q + t(p - q)$  all derivative contributions are nonpositive, and their sum must vanish identically. If  $p \neq q$ , let  $j$  be the largest index with  $p_j > q_j$ . For  $j = n$  the last derivative makes the sum strictly negative. For  $j < n$



vanishing forces  $z_j(t)^2 = z_{j-1}(t)z_{j+1}(t)$  for every  $t$ . But  $z_{j+1}$  is constant, so the right side has degree at most one in  $t$ , whereas the coefficient of  $t^2$  on the left is  $(p_j - q_j)^2 > 0$ . This is impossible.  $\square$

### 3. HYPERSURFACE SECTIONS AND DOMINATION

The hypersurface identity below relates the volume of a valuation to its value on each function. Its two consequences, domination and the final mixed-multiplicity ratio, will be used in both the rigidity and finite-generation arguments.

**Proposition 3.1.** *For  $0 \neq f \in \mathfrak{m}$  and  $v$  of finite discrepancy,*

$$(3.1) \quad \lim_{m \rightarrow \infty} \frac{e(\mathfrak{a}_m(v); R/(f))}{m^{n-1}} = v(f) \operatorname{vol}(v).$$

*Proof.* First assume that  $f$  is irreducible. Put  $s = v(f)$ ,  $H(t) = \ell(R/\mathfrak{a}_t(v))$ , and  $L(t) = \ell(R/(\mathfrak{a}_t(v) + (f)))$ . Multiplicativity of  $v$  gives the exact colon identity  $(\mathfrak{a}_t(v) : f) = \mathfrak{a}_{t-s}(v)$ . Thus multiplication by  $f$  gives

$$0 \longrightarrow R/\mathfrak{a}_{t-s}(v) \xrightarrow{f} R/\mathfrak{a}_t(v) \longrightarrow R/(\mathfrak{a}_t(v) + (f)) \longrightarrow 0,$$

and consequently  $L(t) = H(t) - H(t - s)$ . The ring  $S = R/(f)$  is reduced and excellent. Theorem 2.2 applied to  $\mathfrak{a}_m(v)S$  supplies a number  $B$  such that

$$L(m) = \frac{B}{(n-1)!} m^{n-1} + o(m^{n-1}), \quad B = \lim_{m \rightarrow \infty} \frac{e(\mathfrak{a}_m(v); S)}{m^{n-1}}.$$

Both  $L$  and  $H$  are nondecreasing, so comparison with integer floors and ceilings extends their asymptotic formulas to real arguments. Since  $H(0) = 0$ , telescoping gives

$$\begin{aligned} H(ks) &= \sum_{i=1}^k L(is) \\ &= \frac{Bs^{n-1}}{(n-1)!} \sum_{i=1}^k i^{n-1} + o(k^n) = \frac{Bs^{n-1}}{n!} k^n + o(k^n). \end{aligned}$$

On the other hand, the definition of volume gives  $H(ks) = \operatorname{vol}(v) s^n k^n / n! + o(k^n)$ . Hence  $B = s \operatorname{vol}(v)$ .

For general  $f$ , factor  $f = c \prod_i f_i^{r_i}$  in the regular local unique factorization domain  $R$ , where  $c$  is a unit and the  $f_i$  are pairwise nonassociate irreducibles. The associativity formula for multiplicity [10, Chapter 11] gives

$$e(\mathfrak{a}_m(v); R/(f)) = \sum_i r_i e(\mathfrak{a}_m(v); R/(f_i)).$$

Indeed the generic length along  $(f_i)$  is  $r_i$ . Apply the irreducible case and use  $v(f) = \sum_i r_i v(f_i)$ .  $\square$

**Corollary 3.2.** *If  $w \geq v$  and both discrepancies are finite, then, for all  $0 \neq f \in \mathfrak{m}$ ,*

$$(3.2) \quad w(f) \operatorname{vol}(w) \leq v(f) \operatorname{vol}(v).$$

*In particular,  $w \geq v$  and  $\operatorname{vol}(w) = \operatorname{vol}(v)$  imply  $w = v$ .*

*Proof.* The containment  $\mathfrak{a}_m(v) \subseteq \mathfrak{a}_m(w)$  remains true on  $R/(f)$ . Monotonicity of Hilbert–Samuel multiplicity and Proposition 3.1 give (3.2). Volume is positive by Lemma 2.3. When the volumes coincide, (3.2) gives  $w(f) \leq v(f)$  for every  $f$ , so domination gives equality.  $\square$

**Corollary 3.3.** *One has  $b_{n-1}(v) = v(\mathfrak{m}) \operatorname{vol}(v)$ .*

*Proof.* Choose a linear form  $h$  which is sufficiently general for the mixed-multiplicity section formula for every  $\mathfrak{a}_m(v)$ , and such that  $v(h) = v(\mathfrak{m})$ . The latter condition excludes a proper linear subspace of the space of linear forms. Then  $e_{n-1}(\mathfrak{a}_m(v)) = e(\mathfrak{a}_m(v); R/(h))$  for all  $m$ . Proposition 3.1 gives the conclusion.  $\square$

#### 4. DIVISORIAL RIGIDITY AND EQUALITY IDEALS

We prove Theorem 1.2. Throughout this section a divisorial valuation is first normalized as  $v = \operatorname{ord}_E$ , so its value group is  $\mathbb{Z}$ . Positive rescaling does not affect the assertion of coordinate monomiality. Geometric constructions are performed on a smooth algebraic representative of the germ. Completion will be used only for the coordinate classification.

**4.1. Integral closures and the threshold-computing divisor.** We first record a consequence of numerical monotonicity and the multiplicity theorem of Rees. Recall that the latter says that, for nested primary ideals  $K \subseteq L$  in a formally equidimensional local ring,  $e(K) = e(L)$  implies  $\overline{K} = \overline{L}$ ; see [10, Chapter 11]. All rings to which we apply it are regular.

**Lemma 4.1** (The valuation ideals of an equality ideal). *Let  $I$  be  $\mathfrak{m}$ -primary with  $\operatorname{DP}(I) = \operatorname{lct}(I)$ , and let  $\operatorname{ord}_F$  compute  $\operatorname{lct}(I)$ . Set  $q = \operatorname{ord}_F(I)$  and  $w = q^{-1} \operatorname{ord}_F$ . Then*

$$(4.1) \quad \overline{I^m} = \mathfrak{a}_m(w) \quad (m \geq 1), \quad \operatorname{vol}(w) = e(I), \quad \operatorname{DP}(w) = A(w).$$

*In particular, the normalized divisorial valuation computing  $\operatorname{lct}(I)$  is unique.*

*Proof.* Since  $I$  is  $\mathfrak{m}$ -primary, every divisor computing its threshold has center  $o$ , and  $q > 0$ . Blum’s results [5, Theorem 1.3 and Proposition 1.5] imply that the valuation-ideal algebra of  $\operatorname{ord}_F$  is finitely generated: its asymptotic threshold is  $A(\operatorname{ord}_F)$ , strictly larger than  $A(\operatorname{ord}_F) - 1$ . Consequently the integer-indexed algebra for  $w = \operatorname{ord}_F/q$  is also finitely generated; it is the  $q$ -th Veronese of that algebra. There is an integer  $M > 0$  such that

$$\mathfrak{a}_{kM}(w) = \mathfrak{a}_M(w)^k \quad (k \geq 1).$$

Here and below, choosing such a Veronese means choosing degrees in which the graded algebra is generated by its first piece.

The normalization gives  $w(I) = 1$ , whence  $I^m \subseteq \mathfrak{a}_m(w)$ . Monotonicity of mixed multiplicities therefore gives  $b_j(w) \leq e_j(I)$  for every  $j$ . Both sequences are positive and log-convex. Applying Lemma 2.5, and then the DP inequality for valuation ideals, yields

$$(4.2) \quad \text{DP}(I) \leq \text{DP}(w) \leq \text{lct}(\mathfrak{a}_\bullet(w)) \leq A(w) = \text{lct}(I).$$

The two ends coincide by hypothesis. Thus every inequality is an equality. The strict negativity of the last derivative in Lemma 2.5 forces  $b_n(w) = e_n(I)$ : if the last coordinates differed, integrating that derivative along the segment would make the first inequality strict. Hence  $\text{vol}(w) = e(I)$ .

For every positive integer  $k$  we have

$$I^{kM} \subseteq \mathfrak{a}_{kM}(w), \quad e(I^{kM}) = (kM)^n e(I) = e(\mathfrak{a}_{kM}(w)).$$

The last equality follows from the generated Veronese and the definition of  $b_n(w)$ . A valuation ideal is integrally closed: an integral equation for an element of smaller value would have its leading power as its unique term of least value, which is impossible. Rees's theorem consequently gives  $\overline{I^{kM}} = \mathfrak{a}_{kM}(w)$ .

Now fix any  $m \geq 1$ . The inclusion  $\overline{I^m} \subseteq \mathfrak{a}_m(w)$  again follows from integral closedness. Conversely, if  $f \in \mathfrak{a}_m(w)$ , then  $f^M \in \mathfrak{a}_{mM}(w) = \overline{I^{mM}}$ . For every divisorial test valuation  $u$ , the valuative criterion for integral closure gives  $Mu(f) \geq mM u(I)$ . Dividing by  $M$  and applying the same criterion to  $I^m$  shows  $f \in \overline{I^m}$ . This proves (4.1).

If another divisor computes the threshold, normalize it to obtain  $w'$  with  $w'(I) = 1$ . The argument gives  $\mathfrak{a}_m(w') = \overline{I^m} = \mathfrak{a}_m(w)$  for all integers  $m$ . These integer thresholds recover every function value, since

$$w(f) = \lim_{k \rightarrow \infty} \frac{1}{k} \max\{m \in \mathbb{Z}_{\geq 0} : f^k \in \mathfrak{a}_m(w)\}.$$

Thus  $w' = w$ , proving uniqueness.  $\square$

**Lemma 4.2** (Extraction of the equality divisor). *Let  $v = \text{ord}_E$  satisfy  $\text{DP}(v) = A(v)$  in dimension  $n \geq 2$ . Then  $E$  is a Kollár component. Moreover, one can choose an integer  $N > 0$  such that, with  $I = \mathfrak{a}_N(v)$ ,*

$$(4.3) \quad \mathfrak{a}_{kN}(v) = I^k, \quad v(I) = N, \quad e_j(I) = N^j b_j(v), \quad \text{DP}(I) = \text{lct}(I) = \frac{A(v)}{N}.$$

*Proof.* The DP-threshold chain gives  $\text{lct}(\mathfrak{a}_\bullet(v)) = A(v)$ . Blum's finite-generation result applies as in the preceding proof. Choose a generated Veronese, giving the first identity in (4.3). Since  $v(\mathfrak{a}_\bullet(v)) = 1$ , taking the infimum along this cofinal subsequence gives  $v(I)/N = 1$ . Mixed multiplicities of powers scale by  $e_j(I^k) = k^j e_j(I)$ , proving the third identity. Finally, the ordinary DP inequality and evaluation at  $v$  give

$$\frac{A(v)}{N} = \text{DP}(I) \leq \text{lct}(I) \leq \frac{A(v)}{v(I)} = \frac{A(v)}{N}.$$

By Lemma 4.1,  $\text{ord}_E$  is the unique divisorial valuation computing  $\text{lct}(I)$ , up to scaling. For every divisor  $F$  with  $\text{ord}_F(\mathfrak{a}_\bullet(v)) > 0$ , the generated Veronese gives

$$\text{ord}_F(\mathfrak{a}_\bullet(v)) = \frac{\text{ord}_F(I)}{N}.$$

If  $F$  is different from  $E$  as a divisorial valuation, uniqueness therefore implies

$$A(v) < \frac{A(\text{ord}_F)}{\text{ord}_F(\mathfrak{a}_\bullet(v))}.$$

For divisors with zero denominator the right side is interpreted as  $+\infty$ . Blum's plt-extraction criterion [5, Proposition 4.4] now applies. It extracts  $E$  as the only exceptional divisor, with  $(Y, E)$  purely log terminal and  $-(K_Y + E)$  relatively ample. This is precisely the Kollár-component property, since the center is the closed point and  $n \geq 2$ .  $\square$

#### 4.2. Descent to a general hyperplane.

**Lemma 4.3** (Hyperplane descent). *Use the notation of Lemma 4.2. Choose a sufficiently general linear form  $h$ , and put*

$$H = (h = 0), \quad S = R/(h), \quad J = IS, \quad s = v(h) = v(\mathfrak{m}).$$

Then

$$(4.4) \quad \text{DP}(J) = \text{lct}_H(J) =: c_H, \quad A(v) = s + Nc_H.$$

*Proof.* Choose  $h$  to have value  $s = v(\mathfrak{m})$  and to satisfy the general-section identities  $e_j(J) = e_j(I)$  for  $0 \leq j \leq n-1$ . Here the mixed multiplicities of  $J$  are computed with the maximal ideal of the  $(n-1)$ -dimensional regular ring  $S$ . The bad choices for the first condition form a proper linear subspace, and the section identities hold on a nonempty open set of choices. Corollary 3.3 and (4.3) give the exact ratio

$$(4.5) \quad \frac{e_{n-1}(I)}{e_n(I)} = \frac{b_{n-1}(v)}{Nb_n(v)} = \frac{s}{N}.$$

The pair  $(H, J^{c_H})$  is log canonical near  $o$ . Inversion of adjunction for the smooth Cartier divisor  $H$  implies that  $(X, H + I^{c_H})$  is log canonical near  $H$ ; we use [9, Theorem 1.1]. In this notation an ideal boundary means that the discrepancy of  $u$  is  $A_X(u) - u(h) - c_H u(I)$ . Evaluating its nonnegativity at  $v$  gives

$$c_H \leq \frac{A(v) - s}{N} = \text{DP}(I) - \frac{e_{n-1}(I)}{e_n(I)} = \text{DP}(J) \leq c_H.$$

The last inequality is the DP inequality on  $H$ ; in dimension one it is the elementary equality between reciprocal order and threshold. All inequalities are equalities. This proves both assertions of (4.4).  $\square$

**4.3. Rees algebras and hyperplane restriction.** The next step concerns the scheme structure of the hyperplane in the degeneration, not just its support. Put  $\mathfrak{a}_m = \mathfrak{a}_m(v)$  for  $m \in \mathbb{Z}$ , with  $\mathfrak{a}_m = R$  for  $m \leq 0$ , and define

$$T = \bigoplus_{m \in \mathbb{Z}} \mathfrak{a}_m t^{-m} \subset R[t, t^{-1}], \quad G = T/(t) = \text{gr}_v R,$$

$$\mathfrak{b}_m = (\mathfrak{a}_m + (h))/(h) \subset S, \quad T_H = \bigoplus_{m \in \mathbb{Z}} \mathfrak{b}_m t^{-m} \subset S[t, t^{-1}].$$

These are extended Rees algebras: allowing negative indices puts  $t$  in the algebra, and setting  $t = 0$  produces the associated graded ring. They are finitely generated by Lemma 4.2 and passage to a quotient. Let  $h_0 = \text{in}_v(h) \in G_s$ .

**Lemma 4.4** (The quotient algebra). *With the preceding notation,*

$$(4.6) \quad T/(ht^{-s}) \simeq T_H, \quad T_H/(t) \simeq G/(h_0), \quad \mathfrak{b}_{kN} = J^k \quad (k \geq 1).$$

Moreover,  $T_H$  is a domain and  $t$  is a non-zero divisor on  $T_H$ .

*Proof.* For every integer  $m$ , the valuation identity gives

$$(4.7) \quad \mathfrak{a}_m \cap hR = h\mathfrak{a}_{m-s}.$$

Indeed,  $hg \in \mathfrak{a}_m$  if and only if  $s + v(g) \geq m$ . The natural graded surjection  $T \rightarrow T_H$  has in degree  $m$  exactly the kernel in (4.7). This kernel is the degree- $m$  part of the principal ideal  $(ht^{-s})$ . This proves the first isomorphism; setting  $t = 0$  proves the second. The identity  $\mathfrak{b}_{kN} = J^k$  follows by reducing  $\mathfrak{a}_{kN} = I^k$  modulo  $h$ . Since  $T_H$  is a subring of the domain  $S[t, t^{-1}]$ , it is a domain. In particular  $t$  is a non-zero divisor on it.  $\square$

#### 4.4. Normality and reducedness of the hyperplane degeneration.

Write  $\mathcal{X} = \text{Spec } T$ ,  $X_0 = \text{Spec } G$ , and  $\mathcal{H} = V(ht^{-s}) \subset \mathcal{X}$ . The divisor  $\mathcal{H}$  is Cartier and prime, and after inverting  $t$  it is  $H \times \mathbb{G}_m$ . The preceding domain assertion shows it has no component or embedded part supported over  $t = 0$ . Thus it is exactly the scheme-theoretic closure of  $H \times \mathbb{G}_m$ .

**Lemma 4.5** (Compatibility with the hyperplane). *The pair  $(\mathcal{X}, X_0)$  is purely log terminal, and*

$$(4.8) \quad (\mathcal{X}, X_0 + \mathcal{H}) \text{ is log canonical.}$$

*Proof.* On a common log resolution of  $(X, H, I)$ , write  $I\mathcal{O}_Y = \mathcal{O}_Y(-F)$  and include  $E$  among the resolved divisors. Choose an integer  $r > c_H$  and sufficiently general elements  $f_1, \dots, f_r$  of a finite-dimensional generating space for  $I$ . Set

$$D = \frac{c_H}{r} \sum_{i=1}^r \text{div}(f_i), \quad \Gamma = H + D.$$

The moving parts of these divisors on  $Y$  are base-point-free. Bertini's theorem lets us choose them smooth and transverse to each other and to the fixed resolution divisor. Their coefficients are  $c_H/r < 1$ , while their fixed part is

exactly  $c_H F$ . Consequently the log-canonicity inequalities for  $(X, H + I^{c_H})$  imply those for  $(X, \Gamma)$ . Also  $v(f_i) = N$  for general  $f_i$ , so

$$A_{X, \Gamma}(v) = A(v) - s - c_H N = 0.$$

After shrinking the smooth germ,  $K_X$  is trivial and all the divisors just chosen are principal. Hence  $K_X + \Gamma \sim_{\mathbb{Q}} 0$  locally,  $(X, \Gamma)$  is log canonical, and  $o$  is an lc center. Thus  $\Gamma$  is a local  $\mathbb{Q}$ -complement with  $E$  as an lc place.

The Kollár-component degeneration theorem [17, Lemma 2.17] now gives

$$(4.9) \quad (\mathcal{X}, X_0) \text{ is plt}, \quad (\mathcal{X}, X_0 + \Gamma_{\mathcal{X}}) \text{ is lc},$$

where  $\Gamma_{\mathcal{X}}$  is the closure of the boundary on the nonzero fibers. Its coefficient along  $\mathcal{H}$  is one. Removing the other effective boundary components yields (4.8).  $\square$

**Proposition 4.6** (Normal quotient and reduced central fiber). *The domain  $T_H$  is normal, and  $G/(h_0)$  is reduced.*

*Proof.* For an exceptional divisor  $F'$  over  $\mathcal{X}$  whose center is contained in  $X_0$ , removing  $X_0 = \text{div}(t)$  from (4.8) gives

$$A_{\mathcal{X}, \mathcal{H}}(F') = A_{\mathcal{X}, X_0 + \mathcal{H}}(F') + \text{ord}_{F'}(t) > 0.$$

If the center is not contained in  $X_0$ , its generic point lies in  $\mathcal{X} \setminus X_0 = X \times \mathbb{G}_m$ . There the pair is the product of the smooth pair  $(X, H)$  with  $\mathbb{G}_m$ , and its exceptional discrepancies are positive. Thus  $(\mathcal{X}, \mathcal{H})$  is plt. A coefficient-one prime divisor in a plt pair is normal, so  $T_H$  is normal. This standard normality statement and adjunction for plt pairs are reviewed in [12, Chapter 5].

Set  $C = G/(h_0)$ . Lemma 4.5 implies that  $X_0$  is normal and klt by adjunction, hence  $G$  is Cohen–Macaulay [12, Chapter 5]. Here  $X_0$  is Cartier; its different is zero at every codimension-one point, as can also be seen from the following local calculation. Let  $\eta$  be a codimension-one point of  $X_0$  at which  $h_0$  vanishes. The ring  $\mathcal{O}_{X_0, \eta}$  is a discrete valuation ring. The ambient local ring  $B = \mathcal{O}_{\mathcal{X}, \eta}$  has dimension two, and  $B/(t) = \mathcal{O}_{X_0, \eta}$ . Lifting a generator of the maximal ideal of this discrete valuation ring shows that the maximal ideal of  $B$  is generated by two elements, one of them  $t$ . Thus  $B$  is regular.

Ordinary adjunction for the lc pair (4.8), at this regular point of the ambient space, says that the coefficient of  $\mathcal{H}|_{X_0}$  at  $\eta$  is at most one. That coefficient is precisely  $\text{ord}_{\eta}(h_0)$ , a positive integer. It is therefore one. Every irreducible component of  $C$  consequently has multiplicity one. Finally,  $h_0$  is a non-zero divisor in the domain  $G$ , so  $C$  is Cohen–Macaulay and has no embedded associated primes. A Noetherian ring with no embedded associated primes which is reduced at its minimal primes is reduced. This proves the claim.  $\square$

**4.5. Simultaneous induction and lifting of parameters.** We now prove the two statements of Theorem 1.2 together. In dimension one a centered divisorial valuation is a multiple of ordinary order, and every primary ideal is

a power of the maximal ideal. Both assertions follow immediately. Suppose the ideal assertion is known in dimension  $n - 1$ , and let  $v = \text{ord}_E$  be an equality valuation in dimension  $n$ .

Use the notation and results of Lemmas 4.2–4.5 and Proposition 4.6. We pass to the completions of  $R$  and  $S$  at this point. This passage preserves every finite-colength ideal quotient and its associated graded piece. It also preserves the normality of  $T_H$ : the completion map of the excellent regular local ring  $S$  is a regular morphism, and its base change to the finite-type normal  $S$ -algebra  $T_H$  is normal; see [15, Tag 0BFK]. The base-changed algebra embeds in  $\widehat{S}[t, t^{-1}]$  by flatness. We continue to write  $T_H$ ,  $\mathfrak{b}_m$ , and  $J$  for these extensions.

The finite-discrepancy valuation itself extends uniquely to  $\widehat{R}$ . If  $f \neq 0$  has order  $k$ , its sufficiently long polynomial truncations have order  $k$  and values at most  $A(v)k$ , whereas their successive differences have arbitrarily large value by Lemma 2.1. Their values therefore stabilize. This defines the extension, and the same argument preserves products, sums, valuation ideals, and initial forms.

**Lemma 4.7** (Identification of the quotient filtration). *Under the dimension- $(n - 1)$  ideal assertion, there are formal regular parameters  $y_1, \dots, y_{n-1}$  of  $\widehat{S}$  and positive integers  $a_1, \dots, a_{n-1}$  such that*

$$(4.10) \quad \mathfrak{b}_m = \{g : w(g) \geq m\} \quad (m \geq 0), \quad w(y_i) = a_i,$$

where  $w$  is the indicated monomial valuation. In particular,  $G/(h_0) \simeq \mathbb{C}[Y_1, \dots, Y_{n-1}]$ , with  $\deg Y_i = a_i$ .

*Proof.* First every power  $J^k = \mathfrak{b}_{kN}$  is integrally closed. Indeed, if  $g$  is integral over  $J^k$ , an integral equation for  $g$ , multiplied by the corresponding power of  $t^{-kN}$ , shows that  $gt^{-kN}$  is integral over  $T_H$ . It lies in the fraction field of  $T_H$ , since  $\widehat{S}$  and  $t$  lie in  $T_H$ . Normality gives  $gt^{-kN} \in T_H$ , hence  $g \in J^k$ .

By Lemma 4.3,  $J$  is an equality ideal. The induction hypothesis and integral closedness give formal regular parameters  $y_i$  and positive integers  $d_i$  such that

$$J = \overline{(y_1^{d_1}, \dots, y_{n-1}^{d_{n-1}})}.$$

Let  $w$  be the rational monomial valuation with  $w(y_i) = N/d_i$ . The Newton polyhedron of this pure-power ideal is  $\{u \geq 0 : \sum_i u_i/d_i \geq 1\}$ . Integral closure of its  $k$ -th power is therefore  $\{g : w(g) \geq kN\}$ . Since  $J^k$  is integrally closed and has the same integral closure as that power, we obtain

$$(4.11) \quad J^k = \{g : w(g) \geq kN\} \quad (k \geq 1).$$

We must identify the intermediate filtration levels, not just this Veronese. For  $m \geq 0$  and  $g \in \widehat{S}$ ,

$$(4.12) \quad g \in \mathfrak{b}_m \iff g^N \in \mathfrak{b}_{mN} = J^m.$$

The forward direction uses multiplicativity of the filtration. For the reverse,  $(gt^{-m})^N \in T_H$ , so  $gt^{-m}$  is integral over the normal domain  $T_H$  and belongs to it. Combining (4.12) with (4.11) yields  $\mathfrak{b}_m = \{g : w(g) \geq m\}$  for every  $m$ .

The numbers  $N/d_i$  are in fact integers. If  $r = w(y_i)$  were not integral, set  $k = \lfloor r \rfloor$ . The class of  $y_i$  in  $\mathfrak{b}_k/\mathfrak{b}_{k+1}$  is nonzero. Choose an integer  $p > 0$  with  $p(r - k) \geq 1$ . Then  $y_i^p \in \mathfrak{b}_{pk+1}$ , so this nonzero class is nilpotent in  $G/(h_0)$ . This contradicts its reducedness from Proposition 4.6. Thus  $a_i = N/d_i$  are positive integers. The integer-indexed graded ring of this monomial valuation is the asserted weighted polynomial ring.  $\square$

*Divisorial induction step.* Let  $y_1, \dots, y_{n-1}$  be the parameters in Lemma 4.7. Choose a lift  $z_i \in \mathfrak{a}_{a_i}(v)\widehat{R}$  of each  $y_i$ . Its value is exactly  $a_i$ , since a higher value would put  $y_i$  in  $\mathfrak{b}_{a_i+1}$ . The elements

$$z_1, \dots, z_{n-1}, h$$

form a regular system of parameters of  $\widehat{R}$ : their images in  $\widehat{R}/(h)$  are regular parameters, and  $h$  has ordinary order one.

Their initial forms generate  $G$ . Indeed, take a homogeneous element  $g \in G$  of degree  $d$ . Modulo  $h_0$ , express it as a weighted homogeneous polynomial in the images of the  $\text{in}_v(z_i)$ . Subtract that polynomial from  $g$ . The remainder is  $h_0 g_1$  for a homogeneous element  $g_1$  of degree  $d - s$ . Since  $s > 0$ , induction on  $d$  expresses  $g_1$ , and then  $g$ , in the proposed generators. The induction starts in degree zero, where  $G_0 = \mathbb{C}$ .

We obtain a graded surjection

$$\mathbb{C}[Z_1, \dots, Z_{n-1}, U] \longrightarrow G, \quad \deg Z_i = a_i, \quad \deg U = s.$$

The target is a domain of dimension  $n$ . For example, its dimension follows from the degree- $n$  cumulative Hilbert growth  $\ell(R/\mathfrak{a}_m(v)) \sim \text{vol}(v)m^n/n!$  and finite generation. The kernel is a prime ideal of height zero in a polynomial ring, and so is zero. Thus no nonzero polynomial in these initial forms can cancel. For any formal series in these parameters, there are only finitely many monomials of bounded weight, and its least-weight part has nonzero initial form. Its value is therefore the minimum of the weights of its monomials. This is exactly coordinate monomiality of  $v$ .  $\square$

*Ideal induction step.* Let  $K$  be an  $\mathfrak{m}$ -primary equality ideal and choose a divisor  $v = \text{ord}_E$  computing its threshold. Put  $q = v(K)$ . Lemma 4.1 gives  $\text{DP}(v) = A(v)$  and

$$(4.13) \quad \overline{K} = \mathfrak{a}_q(v), \quad e(K) = q^n \text{vol}(v).$$

The divisorial assertion just proved gives formal regular parameters  $z_i$  with positive integral weights  $\alpha_i = v(z_i)$ . Hence  $\text{vol}(v) = 1/(\alpha_1 \cdots \alpha_n)$ . In these parameters  $\overline{K}$  is the monomial ideal generated by the monomials with  $\sum_i \alpha_i u_i \geq q$ .

Let  $P$  be its Newton polyhedron and let  $Q = \{u \in \mathbb{R}_{\geq 0}^n : \sum_i \alpha_i u_i \geq q\}$ . Then  $P \subseteq Q$ . The multiplicity formula for a primary monomial ideal, together



with (4.13), gives

$$n! \operatorname{Vol}(\mathbb{R}_{\geq 0}^n \setminus P) = \frac{q^n}{\alpha_1 \cdots \alpha_n} = n! \operatorname{Vol}(\mathbb{R}_{\geq 0}^n \setminus Q).$$

The Newton-polyhedron description of integral closure and this multiplicity formula are recalled in [3, 10]. The closed convex polyhedra  $P$  and  $Q$  must coincide. Indeed, if an axial vertex  $(q/\alpha_i)e_i$  of  $Q$  were outside  $P$ , closedness would give a small neighborhood of it disjoint from  $P$  whose intersection with  $Q$  has positive volume, contradicting the displayed equality. If all axial vertices belong to  $P$ , convexity and stability under addition of the positive orthant give  $Q \subseteq P$  directly.

Every vertex of the Newton polyhedron of a monomial ideal is an integral exponent vector of a monomial generator. Thus  $d_i = q/\alpha_i$  is a positive integer for every  $i$ . The identity  $P = Q$  proves  $\overline{K} = (z_1^{d_1}, \dots, z_n^{d_n})$  in  $\hat{R}$ .

Conversely, order the positive integers so that  $d_1 \leq \dots \leq d_n$ . The pure-power ideal has

$$e_j = d_1 \cdots d_j \quad (1 \leq j \leq n), \quad \operatorname{lct} = \sum_{i=1}^n \frac{1}{d_i}.$$

These are the usual mixed-multiplicity and threshold formulas for a diagonal ideal [3]. Hence  $\operatorname{DP} = \operatorname{lct}$ . Integral closure, formal changes of regular parameters, and completion preserve the invariants in this equality, giving the converse for  $K$ .

Finally, an  $\hat{\mathfrak{m}}$ -primary ideal of  $\hat{R}$  contains  $\hat{\mathfrak{m}}^L$  for some  $L$ . Truncating a finite generating set modulo  $\hat{\mathfrak{m}}^L$  and adjoining the degree- $L$  monomials expresses it as the extension of an  $\mathfrak{m}$ -primary ideal in  $R$ . Its mixed multiplicities and threshold agree with those of this algebraic ideal (a log resolution remains a log resolution after completion). The assertion therefore holds for formal ideals as well. This completes the simultaneous induction and the proof of Theorem 1.2.  $\square$

*Remark 4.8.* The argument uses discreteness and a generated Veronese at essential points: the integral parameter  $N$ , the root test (4.12), and induction on integer degrees. Finite generation of an associated graded ring for a general real valuation does not by itself provide these properties. Thus this proof does not establish the arbitrary-real-valuation version of coordinate monomiality.

## 5. EQUALITY IDEALS IN ARBITRARY CODIMENSION

In this section  $S = \mathbb{C}[[x_1, \dots, x_n]]$  and  $\mathfrak{n}$  is its maximal ideal. We prove Theorem 1.1. The reduction to primary ideals is separated from the primary classification itself: the numerical and formal approximation lemmas below do not use Section 4.

### 5.1. Transverse primary truncations.

**Lemma 5.1** (Mixed multiplicities of a transverse truncation). *Let  $0 \neq I \subsetneq S$  have height at least  $l < n$ , set  $s_j = \sigma_j(I)$  for  $0 \leq j \leq l$ , and put  $r = n - l$ . Choose  $f_1, \dots, f_l \in I$  and independent linear forms  $z_1, \dots, z_r$  as in (2.1). Write  $B = (z_1, \dots, z_r)$  and  $K_N = I + B^N$ . Then  $K_N$  is primary to  $\mathfrak{n}$  and*

$$(5.1) \quad e_j(K_N) \leq p_j(N), \quad p_j(N) = \begin{cases} s_j, & 0 \leq j \leq l, \\ s_l N^{j-l}, & l < j \leq n. \end{cases}$$

For every integer  $N \geq s_l/s_{l-1}$ ,

$$(5.2) \quad E_n(K_N) \geq E_l(I) + \frac{r}{N}.$$

*Proof.* The radical of  $I + B^N$  is the radical of  $I + B$ , which is  $\mathfrak{n}$  by (2.1). For  $j \leq l$ , monotonicity of the extended mixed multiplicity gives  $e_j(K_N) \leq \sigma_j(I) = s_j$ .

For  $j = l + q$ , where  $0 \leq q \leq r$ , consider the parameter tuple

$$f_1, \dots, f_l, z_1^N, \dots, z_q^N, z_{q+1}, \dots, z_r.$$

Its first  $j$  elements belong to  $K_N$  and its last  $n - j$  elements belong to  $\mathfrak{n}$ . The mixed-multiplicity comparison recalled in Section 2 therefore gives

$$\begin{aligned} e_j(K_N) &\leq \ell(S/(f_1, \dots, f_l, z_1^N, \dots, z_q^N, z_{q+1}, \dots, z_r)) \\ &= N^q \ell(S/(f_1, \dots, f_l, z_1, \dots, z_r)) = N^q s_l. \end{aligned}$$

For clarity, this comparison does not require the displayed tuple to be general for  $K_N$ . Apply monotonicity to the principal ideals generated by its entries. Their extended mixed multiplicity is the colength of the parameter ideal, because  $S$  is regular. This proves (5.1).

The ratios  $s_j/s_{j-1}$  are nondecreasing by log convexity. Appending the constant ratio  $N \geq s_l/s_{l-1}$  thus makes  $p(N)$  a positive log-convex vector. The vector of mixed multiplicities of  $K_N$  is also positive and log-convex, and is coordinatewise bounded by  $p(N)$ . Lemma 2.5 gives

$$E_n(K_N) \geq F(p(N)) = \sum_{j=0}^{l-1} \frac{s_j}{s_{j+1}} + \frac{r}{N}.$$

□

**Proposition 5.2** (Persistence of equality under truncation). *In the notation of Lemma 5.1, assume  $\text{lct}(I) = E_l(I)$ . Then  $\text{ht } I = l$  and, for every integer  $N \geq s_l/s_{l-1}$ ,*

$$(5.3) \quad \text{lct}(K_N) = E_n(K_N) = E_l(I) + \frac{r}{N}, \quad e_j(K_N) = p_j(N) \quad (0 \leq j \leq n).$$

*Proof.* If  $\text{ht } I \geq l + 1$ , the general-section DP inequality gives  $\text{lct}(I) \geq E_{l+1}(I) > E_l(I)$ , a contradiction. By (2.2),

$$\text{lct}(K_N) \leq \text{lct}(I) + \text{lct}(B^N) = E_l(I) + \frac{r}{N}.$$

Combine this with Lemma 5.1 and the primary DP inequality to obtain

$$E_l(I) + \frac{r}{N} \leq E_n(K_N) \leq \text{lct}(K_N) \leq E_l(I) + \frac{r}{N}.$$

All terms coincide. The equality assertion in Lemma 2.5, applied to the comparable vectors  $(e_j(K_N))$  and  $p(N)$ , gives equality of every coordinate.  $\square$

## 5.2. Integral closure and the adic limit.

**Lemma 5.3** (Continuity of integral closure). *For any proper ideals  $I, B \subset S$ , with  $I \neq 0$ , one has*

$$(5.4) \quad \bigcap_{N \geq 1} \overline{I + B^N} = \bar{I}.$$

Moreover, for every  $k \geq 1$  there is  $N(k)$  such that

$$(5.5) \quad \overline{I + B^N} + \mathfrak{n}^k = \bar{I} + \mathfrak{n}^k \quad (N \geq N(k)).$$

*Proof.* One inclusion in (5.4) is immediate. For the other, suppose  $f \in \overline{I + B^N}$  for every  $N$ . The Briançon–Skoda theorem in a regular local ring of dimension  $n$  gives, for all  $m, N \geq 1$ ,

$$f^{m+n-1} \in \overline{(I + B^N)^{m+n-1}} \subseteq (I + B^N)^m \subseteq I^m + \mathfrak{n}^N.$$

See [10, Chapter 13]. Krull intersection, for fixed  $m$ , gives  $f^{m+n-1} \in I^m$ . If  $u$  is any Rees valuation of  $I$ , then  $(m+n-1)u(f) \geq mu(I)$ . Dividing by  $m$  and letting  $m \rightarrow \infty$  gives  $u(f) \geq u(I)$ . The Rees-valuation criterion for integral closure proves  $f \in \bar{I}$ .

We include the completeness argument needed for (5.5). In a complete Noetherian local ring  $(A, \mathfrak{p})$  with residue field  $\mathbb{C}$ , a decreasing sequence of ideals  $L_N$  with intersection zero is eventually contained in each  $\mathfrak{p}^k$ . Indeed, the images of  $L_N$  in  $A/\mathfrak{p}^k$  stabilize, since this is a finite-dimensional vector space. Call the stable image  $V_k$ . The maps  $V_{k+1} \rightarrow V_k$  are surjective: choose  $N$  large enough for both images to have stabilized and lift using an element of  $L_N$ . If some  $V_k$  were nonzero, these surjections would give a compatible sequence with a nonzero  $k$ -th component. Completeness would yield a nonzero element of  $A$  lying in  $L_N + \mathfrak{p}^j$  for all  $N, j$ . Every ideal is adically closed by Krull intersection, so this element would belong to every  $L_N$ , a contradiction. This proves the assertion, usually called Chevalley's lemma. Apply it to  $A = S/\bar{I}$  and  $L_N = \overline{I + B^N}/\bar{I}$ , using (5.4). The result is (5.5).  $\square$

**Corollary 5.4** (A complete-intersection reduction). *Under Proposition 5.2, the ideal  $J = (f_1, \dots, f_l) \subset I$  satisfies  $\bar{J} = \bar{I}$ . In particular, the analytic spread of  $I$  equals its height  $l$ .*

*Proof.* The parameter ideal  $Q_N = J + (z_1^N, \dots, z_r^N)$  is contained in  $K_N$  and has multiplicity  $N^r s_l$ . Proposition 5.2 gives the same multiplicity for  $K_N$ . Rees's multiplicity theorem yields  $\overline{Q_N} = \overline{K_N}$  for all large  $N$ . Consequently

$I \subseteq \overline{J + B^N}$  for all such  $N$ . Lemma 5.3 gives  $I \subseteq \overline{J}$ , and  $J \subset I$  gives the reverse inclusion of integral closures. The analytic spread is at least the height and at most the number  $l$  of generators of this reduction; see [10, Proposition 8.3.7].  $\square$

**5.3. Compatible coordinate changes.** The following lemma concerns equality of ideals, not merely equality of their radicals. We give the complex algebraic jet-group argument; compare the finite-order equivalence theorem and its proof in [18, Section 5].

**Lemma 5.5** (From finite jets to formal equivalence). *Let  $U, V \subset S$  be ideals. Suppose that, for every  $k \geq 2$ , there is a  $\mathbb{C}$ -algebra automorphism  $\phi_k$  of  $S/\mathfrak{n}^k$  carrying the image of  $U$  onto the image of  $V$ . Then there is a formal  $\mathbb{C}$ -algebra automorphism  $\phi$  of  $S$  with  $\phi(U) = V$ .*

*Proof.* Put  $G_k = \text{Aut}_{\mathbb{C}}(S/\mathfrak{n}^k)$ . These are algebraic groups: coefficients of the images of the variables give coordinates, with invertible linear part. Let  $H_k$  stabilize the image of  $U$ , and let  $T_k$  carry that image onto the image of  $V$ . Then  $H_k$  is a closed algebraic subgroup and  $T_k$  is a nonempty coset of  $H_k$ .

For  $m \geq k$ , truncation maps  $G_m$  to  $G_k$  and  $H_m$  into  $H_k$ . The image of  $H_m$  is closed by the homomorphism theorem for algebraic groups [13, Theorem 1.72]. Thus the images of  $T_m$  in  $G_k$  are nonempty closed cosets. They decrease with  $m$  and stabilize by Noetherianity. Denote the stable image by  $T_k^\infty$ . Choosing  $m$  large enough for stabilization at both levels shows

$$T_{k+1}^\infty \longrightarrow T_k^\infty \quad \text{is surjective.}$$

Choose compatible elements of these sets. Their inverse limit defines an automorphism  $\phi$  of  $S$ . For every  $k$  it carries  $U + \mathfrak{n}^k$  onto  $V + \mathfrak{n}^k$ . Krull intersection, applied to both inclusions, gives  $\phi(U) = V$ .  $\square$

**Lemma 5.6** (Jets of the diagonal models). *Let  $d_1, \dots, d_l$  be positive integers,  $l < n$ , and set*

$$D = \overline{(x_1^{d_1}, \dots, x_l^{d_l})}, \quad D_N = \overline{(x_1^{d_1}, \dots, x_l^{d_l}, x_{l+1}^N, \dots, x_n^N)}.$$

*If  $L$  is the least common multiple of the  $d_i$ , then*

$$(5.6) \quad D \subseteq D_N \subseteq D + \mathfrak{n}^{\lceil N/L \rceil}.$$

*Proof.* Integral closures of monomial ideals are described by their Newton polyhedra [10, Chapter 1]. Membership of a monomial  $x^\alpha$  in  $D_N$  means

$$\sum_{i=1}^l \frac{\alpha_i}{d_i} + \frac{1}{N} \sum_{i=l+1}^n \alpha_i \geq 1.$$

If this monomial does not belong to  $D$ , its first sum is strictly less than one. That sum is an integral multiple of  $1/L$ , so it is at most  $1 - 1/L$ . Hence  $\sum_{i=l+1}^n \alpha_i \geq N/L$  and the monomial belongs to  $\mathfrak{n}^{\lceil N/L \rceil}$ . Applying this to the monomial generators proves the second inclusion. The first follows from the defining containment of pure-power ideals.  $\square$

#### 5.4. Reduction to the primary classification.

**Theorem 5.7** (Primary-to-general reduction). *Fix  $n$ . Suppose every  $\mathfrak{n}$ -primary ideal  $K \subset S$  with  $\text{lct}(K) = E_n(K)$  has integral closure equal to that of a pure-power ideal in some regular parameters of  $S$ . Then Theorem 1.1 holds in dimension  $n$ , for every  $1 \leq l \leq n$ .*

*Proof.* The forward implication for  $l = n$  is the hypothesis. Suppose  $l < n$  and  $\text{lct}(I) = E_l(I)$ . Choose  $B$  as in Lemma 5.1. Proposition 5.2 makes  $K_N = I + B^N$  a primary equality ideal for every sufficiently large integer  $N$ .

The primary hypothesis supplies a formal automorphism  $\Phi_N$  and ordered positive integers  $a_{1,N} \leq \dots \leq a_{n,N}$  such that  $\overline{K_N}$  is the image of the integral closure of the corresponding pure-power ideal. For such an ideal,  $e_j = a_{1,N} \cdots a_{j,N}$ ; see [3]. Mixed multiplicities are preserved by integral closure and automorphisms, since automorphisms preserve  $\mathfrak{n}$ . Equation (5.3) therefore forces

$$(5.7) \quad a_{j,N} = d_j := \frac{s_j}{s_{j-1}} \quad (1 \leq j \leq l), \quad a_{j,N} = N \quad (l < j \leq n).$$

In particular the  $d_j$  are positive integers, independent of  $N$ . They are ordered by log convexity. With the notation of Lemma 5.6, we have

$$(5.8) \quad \overline{K_N} = \Phi_N(D_N).$$

Fix  $k \geq 2$ . Choose  $N$  large enough for (5.5) and for  $\lceil N/L \rceil \geq k$ . Lemma 5.6 and the preservation of  $\mathfrak{n}^k$  under  $\Phi_N$  give

$$\begin{aligned} \Phi_N(D) + \mathfrak{n}^k &= \Phi_N(D_N) + \mathfrak{n}^k \\ &= \overline{K_N} + \mathfrak{n}^k = \overline{I} + \mathfrak{n}^k. \end{aligned}$$

Thus  $D$  and  $\overline{I}$  are equivalent at every finite order. The changes  $\Phi_N$  need not themselves form a compatible sequence. Lemma 5.5 replaces their finite-order existence by a compatible sequence and produces a formal automorphism  $\Phi$  with  $\Phi(D) = \overline{I}$ . Setting  $y_i = \Phi(x_i)$  proves (1.2), and (5.7) gives the stated exponents.

Conversely, suppose (1.2) holds with  $d_1 \leq \dots \leq d_l$ . The diagonal formulas give

$$\sigma_j(I) = d_1 \cdots d_j \quad (1 \leq j \leq l), \quad \text{lct}(I) = \sum_{j=1}^l \frac{1}{d_j}.$$

For the multiplicity formula, restrict to a general  $l$ -dimensional section: the images of  $y_1, \dots, y_l$  are regular parameters, and integral dependence is preserved under the quotient map. Apply the primary diagonal formula on that section and its further general sections. The threshold formula follows from the Newton polyhedron, or from the monomial valuation with weights  $1/d_1, \dots, 1/d_l$  and zero weights in the remaining directions. These invariants are unchanged by integral closure. Substitution gives  $E_l(I) = \text{lct}(I)$ , proving the converse.  $\square$

*Proof of Theorem 1.1.* For  $n = 1$  the assertion is the classification of nonzero proper ideals of a discrete valuation ring. For  $n \geq 2$ , Theorem 1.2(i) verifies the primary hypothesis of Theorem 5.7, including arbitrary formal primary ideals. Apply that reduction.  $\square$

*Remark 5.8.* The general-codimension proof does not require coordinate monomiality for real valuations with nonclosed center. It uses only primary ideals in the same ambient dimension. In particular, Theorem 1.1 does not assert the full valuative statement of [2, Conjecture 6.3].

## 6. FINITE GENERATION FOR REAL EQUALITY VALUATIONS

We now prove Theorem 1.3. The argument uses the numerical estimate of Lemma 2.5 and the domination inequality of Corollary 3.2, and does not require the divisorial classification of Section 4.

### 6.1. A uniform discrepancy estimate.

**Proposition 6.1** (Uniform discrepancy gap). *Suppose  $v(\mathfrak{m}) = 1$  and  $\text{DP}(v) = A(v) = a < \infty$ . Set  $V = \text{vol}(v)$  and*

$$(6.1) \quad \delta_v = \min \left\{ \frac{1}{2}, \frac{n^n}{(2a)^{2n-1}V^2} \right\} > 0.$$

*For every centered valuation  $w \geq v$  of finite discrepancy and every  $0 \neq f \in \mathfrak{m}$ , one has*

$$(6.2) \quad A(w) - a \geq \delta_v \left( \frac{w(f)}{v(f)} - 1 \right).$$

*The same  $\delta_v$  works simultaneously for all  $w$  and  $f$ .*

*Proof.* Put  $r = w(f)/v(f) \geq 1$  and  $W = \text{vol}(w) > 0$ . Corollary 3.2 yields  $r \leq V/W$ . Also Lemma 2.1 yields  $w(f) \leq A(w) \text{ord}_o(f) \leq A(w)v(f)$ , hence  $r \leq A(w)$ .

If  $A(w) \geq 2a$ , then  $A(w) - a \geq A(w)/2 \geq (r-1)/2$ . Suppose next that  $A(w) < 2a$ , and put  $B = 2a$ . By Lemmas 2.3 and 2.4,  $p = (b_j(v))_{j=1}^n$  and  $q = (b_j(w))_{j=1}^n$  satisfy  $q \leq p$ ,  $F(p) = a$ , and  $F(q) = \text{DP}(w) \leq A(w) < B$ . Lemma 2.5 gives

$$A(w) - a \geq \text{DP}(w) - \text{DP}(v) \geq \frac{V - W}{B^{n-1}V^2}.$$

By (2.5),  $W \geq n^n/A(w)^n \geq n^n/B^n$ . Since  $r \leq V/W$ ,

$$V - W = W(V/W - 1) \geq \frac{n^n}{B^n}(r - 1).$$

Combining these inequalities proves the proposition.  $\square$

## 6.2. Threshold rigidity and quasi-monomiality.

**Corollary 6.2** (Uniqueness of the threshold-computing valuation). *Under the hypotheses of Theorem 1.3,  $v$  is, up to scaling, the unique valuation computing  $\text{lct}(\mathbf{a}_\bullet(v))$ , and  $v$  is quasi-monomial.*

*Proof.* Rescaling  $v$  does not change the assertion or the equality  $\text{DP}(v) = A(v)$ , so assume  $v(\mathfrak{m}) = 1$ . By (2.5), the threshold is  $A(v)$ . If  $w$  computes it, rescale  $w$  so that  $w(\mathbf{a}_\bullet(v)) = 1$ . Then  $w \geq v$  by Lemma 2.4, and  $A(w) = A(v)$ . Proposition 6.1 implies  $w(f) = v(f)$  for every nonzero  $f \in \mathfrak{m}$ , so  $w = v$ . Xu's theorem [16, Theorem 1.1] supplies a quasi-monomial valuation computing this ordinary threshold. Uniqueness identifies its ray with that of  $v$ .  $\square$

**6.3. Boundary perturbations and special valuations.** We use the following criterion for special valuations.

**Theorem 6.3** (Special-valuation criterion). *Let  $o \in (X, \Delta)$  be a klt singularity over  $\mathbb{C}$  and let  $v$  be a quasi-monomial valuation centered at  $o$ . Suppose there is  $\delta > 0$  such that, for every effective  $\mathbb{Q}$ -Cartier  $\mathbb{Q}$ -divisor  $H$  with  $v(H) < \delta$ , the germ  $(X, \Delta + H)$  is klt and  $v$  computes*

$$\text{lct}_{(X, \Delta + H)}(\mathbf{a}_\bullet(v)).$$

*Then  $v$  is special,  $\text{gr}_v \mathcal{O}_{X,o}$  is finitely generated, and the induced central pair is klt.*

This is the implication (v) $\Rightarrow$ (iii) in [6, Lemma 3.10], using [17, Theorem 4.1]. More explicitly, a special valuation is represented on a log smooth or toroidal model  $\pi: (Y, E) \rightarrow (X, \Delta)$  as a monomial lc place of a special  $\mathbb{Q}$ -complement  $\Gamma$ . Here  $\Gamma \geq 0$ ,  $(X, \Delta + \Gamma)$  is lc,  $K_X + \Delta + \Gamma \sim_{\mathbb{Q}} 0$  locally, and the strict transform of  $\Gamma$  contains an effective  $\pi$ -ample  $\mathbb{Q}$ -divisor whose support contains no stratum of  $(Y, E)$ . Being an lc place means  $A_{X, \Delta + \Gamma}(v) = 0$ . This positivity on the model is the extra condition beyond merely computing an unperturbed threshold. We apply this criterion to the smooth germ with  $\Delta = 0$ .

**Proposition 6.4** (Stability under small boundaries). *Under the hypotheses and normalization of Proposition 6.1, every effective  $\mathbb{Q}$ -divisor  $H$  with  $v(H) < \delta_v$  satisfies*

$$(6.3) \quad (X, H) \text{ is klt near } o, \quad \text{lct}_{(X, H)}(\mathbf{a}_\bullet(v)) = a - v(H).$$

*Proof.* Work locally at  $o$ . Since  $R$  is regular, some positive multiple  $NH$  is the divisor of an element  $f \in R$ . If  $v(H) = 0$ , the divisor does not pass through  $o$ , and the assertion follows from (2.5). Otherwise  $f \in \mathfrak{m}$ . Put  $h = v(H)$ . Since  $v(\mathfrak{m}) = 1$ ,

$$\text{ord}_o(H) = \text{ord}_o(f)/N \leq h < \delta_v \leq \frac{1}{2}.$$

Skoda's inequality gives  $\text{lct}(H) \geq 1/\text{ord}_o(H) > 1$ ; therefore  $(X, H)$  is klt near  $o$ .

For a valuation  $w$  occurring in the valuative formula for the threshold in (6.3), normalize it by  $w(\mathfrak{a}_\bullet(v)) = 1$ . The linear bounds ensure that this denominator is positive and finite for every centered divisorial valuation; these valuations suffice to compute the infimum. Lemma 2.4 gives  $w \geq v$ . Applying (6.2) to  $f$  gives

$$\begin{aligned} A(w) - a &\geq \delta_v \left( \frac{w(H)}{h} - 1 \right) \\ &\geq h \left( \frac{w(H)}{h} - 1 \right) = w(H) - h. \end{aligned}$$

The second inequality uses both  $h < \delta_v$  and  $w(H) \geq h$ . Thus  $A_{X,H}(w) \geq a - h$  for every normalized valuation in the infimum. The valuation  $v$  itself gives the reverse inequality, so equality holds and  $v$  computes the threshold.  $\square$

*Proof of Theorem 1.3.* The equality  $\text{DP}(v) = A(v)$  is invariant under positive rescaling:  $b_j(cv) = c^{-j}b_j(v)$  and  $A(cv) = cA(v)$ . Normalize  $v(\mathfrak{m}) = 1$ . Corollary 6.2 gives quasi-monomiality, and Proposition 6.4 verifies every hypothesis of Theorem 6.3 with  $\Delta = 0$ . Its conclusion proves the theorem.  $\square$

## 7. THE IDEAL-WEIGHTED VOLUME

Fix  $0 \neq \mathfrak{q} \subseteq R$  and write  $d = \text{ord}_o(\mathfrak{q})$  and  $D = d + 1$ . We first obtain the sharp lower bound for  $F_{\mathfrak{q}}$  and its equality profile, and then apply the preceding results to its minimizers.

### 7.1. The sharp lower bound and attainment.

**Proposition 7.1.** *If  $v(\mathfrak{m}) = 1$  and  $V = \text{vol}(v)$ , then*

$$(7.1) \quad A(v) \geq \text{DP}(v) \geq 1 + (n-1)V^{-1/(n-1)}.$$

*If equality holds between the two ends, then*

$$b_j(v) = V^{j/(n-1)} \quad (1 \leq j < n), \quad \text{DP}(v) = A(v).$$

*Proof.* Corollary 3.3 gives  $b_{n-1}(v) = b_n(v) = V$ . The last summand of  $\text{DP}(v)$  is therefore 1. The product of the first  $n-1$  summands is  $1/V$ , so their sum is at least  $(n-1)V^{-1/(n-1)}$ . Equality in the arithmetic–geometric mean inequality holds exactly when those summands are equal. Starting with  $b_0 = 1$  gives the stated values of  $b_j$ . Equality in the entire chain also gives  $\text{DP}(v) = A(v)$ .  $\square$

*Proof of Theorem 1.4.* The functional is unchanged by replacing  $v$  with  $cv$ , so normalize  $v(\mathfrak{m}) = 1$ . Set  $x = \text{vol}(v)^{-1/(n-1)}$ . As  $\mathfrak{q} \subseteq \mathfrak{m}^d$ , one has  $v(\mathfrak{q}) \geq d$ . Proposition 7.1 gives

$$(7.2) \quad F_{\mathfrak{q}}(v) \geq \frac{(D + (n-1)x)^n}{x^{n-1}} \geq n^n D.$$

The logarithmic derivative of the middle expression is

$$\frac{n(n-1)}{D + (n-1)x} - \frac{n-1}{x} = \frac{(n-1)(x-D)}{x(D + (n-1)x)}.$$



It is negative for  $x < D$ , positive for  $x > D$ , and zero only at  $x = D$ .

To attain the bound, choose  $f \in \mathfrak{q}$  with  $\text{ord}_o(f) = d$ . When  $d = 0$ , take a unit. When  $d > 0$ , choose linear coordinates so that the degree- $d$  initial form of  $f$  does not vanish on the  $x_1$ -axis. Let  $v_*$  be the monomial valuation of weights  $(1, D, \dots, D)$ . Then  $v_*(\mathfrak{q}) = d$ : every element of  $\mathfrak{q}$  has order at least  $d$ , and the pure  $x_1^d$  term of  $f$  has value  $d$ , while no term has smaller value. The standard monomial formulas give

$$A(v_*) = 1 + (n-1)D, \quad \text{vol}(v_*) = D^{-(n-1)}.$$

Thus  $F_{\mathfrak{q}}(v_*) = n^n D$ .

Now let  $v$  be any minimizer. Both inequalities in (7.2) are equalities, so  $x = D$  and  $A(v) + v(\mathfrak{q}) = 1 + (n-1)D + d$ . The separate bounds  $A(v) \geq 1 + (n-1)D$  and  $v(\mathfrak{q}) \geq d$  force equality in each. Proposition 7.1 gives (1.8), including  $\text{DP}(v) = A(v)$ . Theorem 1.3 proves the finite-generation assertion.  $\square$

## 7.2. Divisorial minimizers and nonuniqueness.

**Corollary 7.2** (Divisorial minimizers). *Let  $d = \text{ord}_o(\mathfrak{q})$  and  $D = d + 1$ . A divisorial valuation  $v$ , normalized by  $v(\mathfrak{m}) = 1$ , minimizes  $F_{\mathfrak{q}}$  if and only if  $v(\mathfrak{q}) = d$  and, in some formal regular parameters, its weights are*

$$(1, D, \dots, D).$$

*In particular, the graded ring of every divisorial minimizer is a polynomial ring with these weights.*

*Proof.* For a minimizer, Theorem 1.4 gives the DP equality and the numerical profile (1.8). Theorem 1.2 makes  $v$  monomial in formal parameters. Order its weights as  $\alpha_1 \geq \dots \geq \alpha_n = 1$ . For a monomial valuation the mixed-multiplicity formula is  $b_j(v) = (\alpha_1 \cdots \alpha_j)^{-1}$ ; it follows either from the diagonal-ideal formula or by approximating its simplex valuation ideals. Thus  $b_j(v) = D^{-j}$  for  $j < n$  forces  $\alpha_1 = \dots = \alpha_{n-1} = D$ . The last weight is one by normalization, and  $v(\mathfrak{q}) = d$  is already part of the profile. Conversely, a valuation with these weights and  $v(\mathfrak{q}) = d$  has

$$A(v) = 1 + (n-1)D, \quad \text{vol}(v) = D^{-(n-1)},$$

so  $F_{\mathfrak{q}}(v) = (nD)^n D^{-(n-1)} = n^n D$ . Theorem 1.4 identifies this value as the minimum.  $\square$

*Remark 7.3.* Corollary 7.2 classifies divisorial minimizers; it does not assert that every minimizer is divisorial. For arbitrary minimizers the numerical conditions also determine which ones survive when the ideal is changed: after normalization by  $v(\mathfrak{m}) = 1$ ,

$$\text{Argmin } F_{\mathfrak{q}} = \{v \in \text{Argmin } F_{\mathfrak{m}^d} : v(\mathfrak{q}) = d\}.$$

Indeed  $F_{\mathfrak{q}}(v) \geq F_{\mathfrak{m}^d}(v)$  and both minima equal  $n^n D$ . For example, in dimension two with  $\mathfrak{q} = \mathfrak{m}$ , the two valuations of weights  $(1, 2)$  and  $(2, 1)$

both minimize  $F_q$  and are not proportional. Finite generation does not imply uniqueness of the minimizing ray.

## 8. ANALYTIC SINGULARITIES ATTAINING EQUALITY

Put  $R_{\text{an}} = \mathbb{C}\{x_1, \dots, x_n\}$ , with completion  $\widehat{R} = \mathbb{C}[[x_1, \dots, x_n]]$ . We first prove the analytic counterpart of Theorem 1.1 and then its plurisubharmonic formulation.

### 8.1. Replacing formal parameters by holomorphic coordinates.

**Lemma 8.1** (Analytic descent of a diagonal integral closure). *Let  $0 \neq I \subsetneq R_{\text{an}}$ , and suppose that there are formal regular parameters  $y_1, \dots, y_n$  of  $\widehat{R}$  and positive integers  $d_1, \dots, d_l$  such that*

$$\overline{I\widehat{R}} = \overline{(y_1^{d_1}, \dots, y_l^{d_l})\widehat{R}}.$$

*For every integer  $q \geq 2$ , there are holomorphic coordinates  $z_1, \dots, z_n$  with  $z_i \equiv y_i \pmod{\widehat{\mathfrak{m}}^q}$  and*

$$(8.1) \quad \overline{I} = \overline{(z_1^{d_1}, \dots, z_l^{d_l})R_{\text{an}}}.$$

*Proof.* Choose generators  $f_1, \dots, f_s$  of  $I$ . For a multi-index  $\alpha \in \mathbb{N}^s$ , write  $f^\alpha = \prod_a f_a^{\alpha_a}$ . The integral dependence of  $y_i^{d_i}$  over  $I\widehat{R}$  gives an integer  $N_i \geq 1$  and coefficients  $A_{ik\alpha} \in \widehat{R}$  such that

$$(8.2) \quad y_i^{d_i N_i} + \sum_{k=1}^{N_i} \sum_{|\alpha|=k} A_{ik\alpha} f^\alpha y_i^{d_i(N_i-k)} = 0 \quad (1 \leq i \leq l).$$

Indeed, the coefficient of the  $(N_i - k)$ th power in a monic integral-dependence equation belongs to  $(I\widehat{R})^k$ , which is generated by the finitely many  $f^\alpha$  with  $|\alpha| = k$ . Conversely, each  $f_a$  is integral over  $(y_1^{d_1}, \dots, y_l^{d_l})\widehat{R}$ . Thus there are  $M_a \geq 1$  and  $B_{ak\beta} \in \widehat{R}$  such that

$$(8.3) \quad f_a^{M_a} + \sum_{k=1}^{M_a} \sum_{\substack{\beta \in \mathbb{N}^l \\ |\beta|=k}} B_{ak\beta} \left( \prod_{i=1}^l y_i^{d_i \beta_i} \right) f_a^{M_a-k} = 0 \quad (1 \leq a \leq s).$$

Regard all  $y_i$ ,  $A_{ik\alpha}$  and  $B_{ak\beta}$  as unknown functions of  $x$ . Equations (8.2) and (8.3) form a finite system of polynomial equations in these unknowns with convergent coefficients  $f_a(x)$ . They have the displayed formal solution. The unknowns  $y_{l+1}, \dots, y_n$  are included, although they occur in no equation. Artin's analytic approximation theorem [1, Theorem 1.2] gives a convergent solution agreeing with this formal solution modulo  $\widehat{\mathfrak{m}}^q$ . If some coefficient unknown has a nonzero constant term, subtract that constant before applying the theorem and add it back afterwards.

Denote the resulting coordinate functions by  $z_i$ . Since  $q \geq 2$ , they vanish at  $o$  and have the same invertible linear part as  $y_1, \dots, y_n$ . The holomorphic

inverse function theorem makes them a local coordinate system. The convergent versions of (8.2) imply  $z_i^{d_i} \in \bar{I}$ . The convergent versions of (8.3) imply  $f_a \in \overline{(z_1^{d_1}, \dots, z_l^{d_l})}$ . These two inclusions, followed by taking integral closure, prove (8.1). Only ordinary Artin approximation is used; there are no nested restrictions on the unknown functions.  $\square$

**Corollary 8.2** (Equality ideals in the analytic local ring). *Let  $0 \neq I \subsetneq R_{\text{an}}$  and  $1 \leq l \leq \text{ht } I$ . Then  $\text{lct}(I) = E_l(I)$  if and only if (8.1) holds in holomorphic coordinates for positive integers  $d_1 \leq \dots \leq d_l$ . In this case  $\text{ht } I = l$ ,  $d_j = \sigma_j(I)/\sigma_{j-1}(I)$ , and  $\text{lct}(I) = \sum_{j=1}^l d_j^{-1}$ .*

*Proof.* Completion preserves height, the extended mixed multiplicities and the log canonical threshold. For the mixed multiplicities, apply invariance under completion to each primary ideal  $I + \mathfrak{m}^a$  and then take the defining supremum. For the threshold, the usual log-resolution formula is unchanged after completion: the multiplicities of the pulled-back ideal and the relative canonical divisor are unchanged. More precisely,  $R_{\text{an}}$  is excellent, its completion map is faithfully flat and regular, and [11, Proposition 1.9] applies to this map. Thus equality for  $I$  implies equality for  $I\hat{R}$ . Theorem 1.1, followed by Lemma 8.1, gives (8.1) with convergent coordinates. Conversely, the diagonal ideal in (8.1) has  $\sigma_j = d_1 \cdots d_j$  for  $j \leq l$  and threshold  $\sum_{j=1}^l d_j^{-1}$ . Invariance under integral closure gives the asserted equality and formulas for  $I$ .  $\square$

**8.2. The analytic–algebraic dictionary.** For  $I = (f_1, \dots, f_s)$ , put  $|I| = (\sum_a |f_a|^2)^{1/2}$ . Different finite generating systems give comparable norms on a sufficiently small neighborhood, so  $\log |I|$  is well defined modulo a bounded term.

**Lemma 8.3** (Integral closure and bounded comparison). *If two nonzero ideals  $I, J \subset R_{\text{an}}$  have the same integral closure, then there is  $C \geq 1$  such that*

$$C^{-1}|J| \leq |I| \leq C|J|$$

*near  $o$ . Consequently  $\log |I| = \log |J| + O(1)$  off their common zero set, and their logarithmic singularities have the same type.*

*Proof.* Let  $g$  be integral over  $J$ , with an equation  $g^r + a_1 g^{r-1} + \dots + a_r = 0$  and  $a_k \in J^k$ . Expressing  $a_k$  in products of generators of  $J$  and shrinking the neighborhood bounds all coefficient functions. Hence  $|a_k| \leq C_k |J|^k$  for some constants  $C_k \geq 0$ . At a point where  $|J| > 0$ , set  $t = |g|/|J|$ . The equation gives  $t^r \leq \sum_{k=1}^r C_k t^{r-k}$ . If  $t \geq 1$ , division by  $t^{r-1}$  gives  $t \leq \sum_k C_k$ ; otherwise  $t < 1$  already. At a zero of  $J$  the equation forces  $g = 0$ . Thus  $|g| \leq C_g |J|$  throughout a neighborhood of  $o$ . Apply this argument to every generator of  $I$ , using  $I \subset \bar{J}$ , and combine the finitely many estimates. This gives  $|I| \leq C|J|$ . Interchanging  $I$  and  $J$  proves the other inequality. Taking logarithms off the common zero set proves the final assertion.  $\square$

Let  $\varphi = \lambda \log |I| + O(1)$  be plurisubharmonic and let  $h = \text{codim}_o V(I)$ . For  $1 \leq j \leq h$ , the classical products  $(dd^c \varphi)^j$  and their Lelong numbers at  $o$  are defined in this codimension range. We set

$$(8.4) \quad e_0(\varphi) = 1, \quad e_j(\varphi) = \nu((dd^c \varphi)^j, o), \quad dd^c = \frac{i}{\pi} \partial \bar{\partial}.$$

These are the classical products retaining mass on the singular set, not the non-pluripolar products. When  $h = n$ , in particular,  $e_n(\varphi) = (dd^c \varphi)^n(\{o\})$ . The normalization gives  $(dd^c \log |x|)^n(\{o\}) = 1$ . For these conventions and the codimension restriction, see [8, Section 1] and [14, Section 1 and Theorem 1.4].

**Lemma 8.4** (Threshold and multiplicity correspondence). *With the preceding notation,*

$$(8.5) \quad c_o(\varphi) = \lambda^{-1} \text{lct}(I), \quad e_j(\varphi) = \lambda^j \sigma_j(I) \quad (0 \leq j \leq h).$$

In particular, for  $1 \leq l \leq h$ ,

$$(8.6) \quad \sum_{j=1}^l \frac{e_{j-1}(\varphi)}{e_j(\varphi)} = \lambda^{-1} E_l(I).$$

*Proof.* A bounded additive term multiplies  $e^{-2c\varphi}$  by a function bounded above and below by positive constants. Therefore it does not change local integrability. Since  $e^{-2c\lambda \log |I|} = |I|^{-2c\lambda}$ , the analytic characterization of the log canonical threshold gives the first identity in (8.5).

For the second identity, the comparison theorem for Lelong numbers of Monge–Ampère products shows that boundedly equivalent plurisubharmonic germs have the same  $e_j$  in the stated range; see the comparison used in [8, Lemma 3.1] and [14, Section 2]. Homogeneity contributes the factor  $\lambda^j$ . It remains to identify  $e_j(\log |I|)$ . Choose a sufficiently general  $j$ -dimensional linear section  $L$  through  $o$ . The restriction  $I\mathcal{O}_{L,o}$  is primary, and the generic-section theorem for these Lelong numbers gives

$$e_j(\log |I|) = (dd^c \log |I\mathcal{O}_{L,o}|)^j(\{o\}) = e(I\mathcal{O}_{L,o}).$$

Here the second equality is the standard identification of the residual Monge–Ampère mass of an analytic ideal with its Samuel multiplicity. These two facts are recalled in [14, Section 1 and the proof of Theorem 1.4]. By the general-section description of extended mixed multiplicities, [2, Proposition 2.37], the last term is  $\sigma_j(I)$ . This proves (8.5); taking successive ratios proves (8.6).  $\square$

### 8.3. Classification of analytic singularity types.

*Proof of Corollary 1.5.* By Lemma 8.4, equality (1.9) is equivalent to  $\text{lct}(I) = E_l(I)$ . Corollary 8.2 supplies holomorphic coordinates  $z$  and positive integers  $d_1 \leq \dots \leq d_l$  such that  $\bar{I} = \bar{J}$ , where  $J = (z_1^{d_1}, \dots, z_l^{d_l})$ . Lemma 8.3 now gives

$$\varphi = \frac{\lambda}{2} \log \left( \sum_{j=1}^l |z_j|^{2d_j} \right) + O(1).$$

For nonnegative numbers  $u_1, \dots, u_l$ , one has

$$\max_j u_j \leq \left( \sum_j u_j^2 \right)^{1/2} \leq \sqrt{l} \max_j u_j.$$

Apply this with  $u_j = |z_j|^{d_j}$  and take logarithms to obtain (1.10). Taking radicals in  $\bar{I} = \bar{J}$  identifies the reduced zero set. The formulas for  $\sigma_j(I)$  and  $\text{lct}(I)$  in Corollary 8.2, together with (8.5), prove (1.11).

Conversely, suppose (1.10) holds. The same norm comparison makes  $\varphi$  boundedly equivalent to  $\lambda \log |J|$ . For the diagonal ideal  $J$ ,  $\sigma_j(J) = d_1 \cdots d_j$  and  $\text{lct}(J) = \sum_{j=1}^l d_j^{-1}$ , so Lemma 8.4 gives (1.11). Thus each successive ratio is  $e_{j-1}(\varphi)/e_j(\varphi) = 1/(\lambda d_j)$ , and their sum is  $c_o(\varphi)$ . This proves the reverse implication.  $\square$

*Remark 8.5* (Scope of the classification). The normal form concerns bounded equivalence of singularities; no maximality hypothesis is needed. In the isolated case it therefore applies, in particular, to maximal weights with analytic singularities, without asserting equality of the weights as functions. The effective exponents  $\lambda d_j = e_j(\varphi)/e_{j-1}(\varphi)$  are determined by the mixed Lelong numbers and have rational pairwise ratios. The decomposition into a common scale and integer exponents need not be unique. No classification of arbitrary, non-analytic plurisubharmonic singularities follows from this argument.

### DECLARATION ON THE USE OF ARTIFICIAL INTELLIGENCE

The author selected the research problems. The arguments and exposition were developed with OpenAI Codex and remain subject to the author's mathematical verification.

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